Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Distributed Systems, Profit Intelligence, Unit Economics, Predictive Modeling, Anomaly Detection, Dimensional Data Modeling, Star/Snowflake Schemas, Slowly Changing Dimensions (SCD Type 1/2), Natural Language Processing, Generative AI, Automated QA Testing, Root Cause Analysis, CI/CD, Data Governance, Lineage Tracking

Programming Languages & Tools: Spark, PySpark, Azure Data Factory, Azure Synapse, Azure SQL, Databricks, AWS S3, Redshift, Snowflake, XGBoost, Random Forests, Scikit-Learn, SARIMA, Power BI, Tableau, MicroStrategy, Airflow, One Automation, Python, SQL

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Software Development Engineer - Profit Intelligence & Large-Scale Distributed Data Platforms — **May 2022 – Present**

- **Architected petabyte-scale distributed data processing pipelines** using Apache Spark, PySpark, and Azure Synapse to ingest, process, and crunch hundreds of TBs of incoming financial events and transactional data streams, ensuring 99.9% data availability.
- **Designed scalable predictive machine learning models** utilizing XGBoost, Scikit-Learn, and SARIMA algorithms for customer behavior and demand forecasting, directly optimizing operational unit economics and capacity planning by 20%.
- **Integrated generative AI capabilities and conversational AI frameworks** into internal analytics tools leveraging prompt engineering and automated regex classification to enable natural language querying and cut manual turnaround time by 50%.
- **Engineered automated data quality audit frameworks and governance firewalls** implementing dimensional data modeling (Star/Snowflake schemas), SCD Type 1/2, and anomaly detection systems that reduced operational discrepancies with 95% accuracy.
- **Owned end-to-end production reliability, batch monitoring, and CI/CD operations** across Airflow and cloud data warehouses, authoring rigorous runbooks and conducting Root Cause Analysis (RCA) postmortems to maintain strict SLA standards.

Master of Science in Information Systems & Graduate Certificate in Data Science — **May 2022 – Present**

- **Completed advanced coursework in information systems** and data science at University of Cincinnati, achieving a top-tier cumulative GPA of 3.95/4.0.
- **Focused academic research on scalable data architectures** and applied machine learning algorithms for enterprise data intelligence systems.
- **Applied rigorous statistical methods** to model complex business datasets and simulate predictive financial outcomes.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Graduated with a Bachelor of Engineering** from Pune University MIT, building a rigorous engineering foundation in complex system dynamics.
- **Applied analytical problem-solving methodologies** to model multivariate processes and optimize large-scale operational pipelines.
- **Leveraged quantitative analytics techniques** for systematic data evaluation and performance optimization.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.

- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.